

Ch 07: Tuples and Plotting Sequences with Matplotlib Library

Application Program Development

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Tuples

- **Tuple:** an immutable sequence
 - Very similar to a list
 - Once it is created it cannot be changed
 - Format: 't = (item1, item2)
 - Tuples support operations as lists
 - Subscript indexing for retrieving elements
 - Methods such as `index()`
 - Built in functions such as `len()`, `min()`, `max()`
 - slicing expressions
 - The `in` + and `*` operators
 - Tuples do not support
 - `append()`
 - `remove()`
 - `insert()`
 - `reverse()`
 - `sort()`

Tuple

A Tuple is an immutable sequence, which means that its contents cannot be changed.

Advantages of tuples over lists:

- Processing tuples is faster than processing lists
- Tuples are safe
- Some operations in Python require use of tuples



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Plotting Data with matplotlib Library

- The matplotlib package is a library for creating two-dimensional charts and graphs
- It is **not** part of the standard Python library, so you will have to install it separately (e.g. `pip install`) after you install Python
- The matplotlib package contains a (sub) module named `pyplot` that you will need to import
- Conventionally the following is used to import the `pyplot` module into the `plt` name space:

```
import matplotlib.pyplot as plt
```



Plotting a Line Graph with the plot() Function

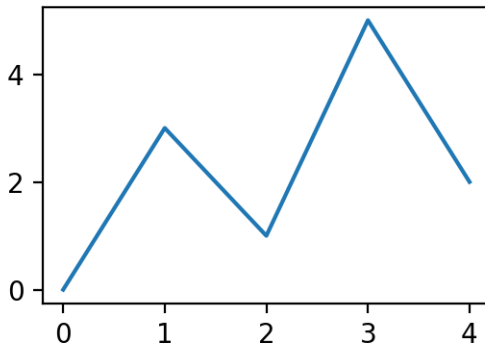
- Use the `plot()` function to create a line graph that connects a series of points with straight lines
 - Each point has an (x, y) coordinate
 - Notice 2 separate sequences of same size, one for x and one for y coordinates

```
import matplotlib.pyplot as plt
```

```
x = [0, 1, 2, 3, 4]
```

```
y = [0, 3, 1, 5, 2]
```

```
plt.plot(x, y)
```



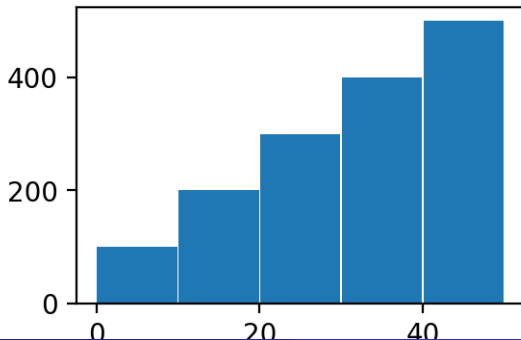
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Plotting a Bar Chart

- Use the `bar()` function in the `matplotlib.pyplot` module to create bar charts
- The function needs two sequences
 - ① one with the x coordinates of each bar's center
 - ② and another with the heights of each bar along the y axis

```
import matplotlib.pyplot as plt

centers = [5, 15, 25, 35, 45]
heights = [100, 200, 300, 400, 500]
bar_width = 9.8
plt.bar(centers, heights, bar_width)
```



Plotting a Pie Chart

- Use the `pie()` function in the `matplotlib.pyplot` module to create a pie chart
- When you call the `pie()` function, you pass a list of values
 - The sum of the values will be used as the value of the whole
 - Each element in the list will become a slice in the pie chart
 - The size of a slice represents the element's value as a percentage of the whole
- The `pie()` function has a `labels` parameter that you can use to display labels for slices
 - A list of strings for the desired labels

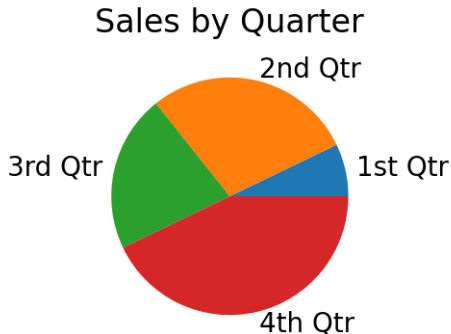
```
import matplotlib.pyplot as plt
```

```
sales = [100, 400, 300, 600]
```

```
labels = ['1st Qtr', '2nd Qtr', '3rd Qtr', '4th Qtr']
```

```
plt.pie(sales, labels=labels)
```

```
plt.title('Sales by Quarter')
```



Summary

- A tuple is an immutable sequence, similar to a list, but it cannot be changed once created
- Has several advantages over a list, and must be used in some situations
- You can use the `matplotlib.pyplot` library module to perform basic plotting
 - Basically plot functions take sequences (lists or tuples) to specify (x,y) coordinates or other plot elements
 - Can plot line graphs, bar charts, pie charts, and many more not shown



References



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